北京大学

硕士研究生学位论文

题目：3/2 波动率模型下的期权定价——使用精确模拟和近似精确模拟方法

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二〇二六 年 五 月

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Option Pricing under the 3/2 Volatility Model: Exact Simulation and Almost Exact Simulation Methods

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ABSTRACT

In the modern workplace, the phenomenon of "shirking" (also known as "moyü" or work avoidance), defined as employees engaging in non-work-related activities or reducing work effort during work hours, is increasingly common. This behavior not only affects organizational efficiency but also reflects profound economic issues within the employment relationship. This paper explores the economic principles behind shirking from the perspective of information economics, based on principal-agent theory and game theory. It investigates how employees choose optimal shirking strategies under conditions of information asymmetry, and how employers design optimal incentive and monitoring mechanisms in response. The research aims to reveal the underlying economic logic of this behavior and provide theoretical insights and (simulated) empirical evidence for understanding and managing modern labor relations.

This study first constructs a concise theoretical model. In this model, a risk-neutral employer cannot directly observe the effort level of a risk-neutral employee but can incentivize the employee by setting a minimum effort standard, paying a fixed wage, and implementing a probabilistic monitoring and punishment mechanism. The model derivation shows that the employee's optimal decision exhibits a threshold effect: the employee chooses to meet the minimum standard only if the expected penalty (monitoring probability multiplied by penalty magnitude) exceeds the cost of achieving that standard; otherwise, the employee chooses zero effort (maximizing shirking). Furthermore, the model analyzes how the employer, balancing output gains, wage payments, and monitoring costs, selects the optimal minimum effort requirement and monitoring probability.

To test the implications of the theoretical model, this paper conducts an empirical analysis using (fictional) panel data from the "China Enterprise Employee Survey (CEES)" and employing a fixed-effects econometric model. The results indicate that increased enterprise monitoring intensity and a higher share of performance pay are significantly negatively cor-

related with the employee shirking index, confirming the effectiveness of monitoring and incentives in curbing opportunistic behavior. Conversely, higher task complexity is positively associated with the shirking index, possibly reflecting the difficulty of effectively monitoring complex tasks.

Integrating theoretical deduction with (fictional) empirical testing, this research deepens the understanding of the economic roots of shirking under information asymmetry. The findings offer implications for managers in designing more effective incentive mechanisms (e.g., balancing performance pay and monitoring investments) and management strategies (e.g., adopting differentiated approaches for tasks of varying complexity). This paper emphasizes that understanding the economics of shirking does not endorse the behavior but aims to promote more harmonious and efficient labor relations through scientific analysis.

KEY WORDS: Option Pricing, 3/2 Volatility Model, Exact Simulation, Almost Exact Simulation, Numerical Methods

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Chapter 1 Introduction

1.1 Research Background

In modern organizational management practices, the phenomenon of employee "shirking" (also referred to as "loafing on the job" or "work avoidance") is increasingly attracting attention. It refers to the behavior where employees, during work hours, do not fully dedicate themselves to tasks directly related to their job responsibilities, but instead engage in non-work activities, reduce their effort level, or procrastinate on work progress. This phenomenon is not merely a matter of individual laziness but is rooted in the complex organizational environment and employment relationship, particularly where information asymmetry commonly exists between the employer (principal) and the employee (agent) (Akerlof, 1970; Spence, 1973). Employers often find it difficult to precisely observe and measure the actual effort level and time investment of employees, which provides employees with the space to selectively exert effort and seek maximization of their personal utility (e.g., pursuing leisure, handling private affairs), thereby engaging in some degree of "shirking" behavior (Alchian et al., 1972).

Although the phenomenon of "shirking" is widespread, its underlying economic principles, particularly how employees make "optimal" shirking decisions in an environment of asymmetric information, and how these decisions interact with corporate incentive mechanisms and monitoring strategies, have not yet been fully and systematically explored. Existing literature in management and organizational behavior often analyzes it from psychological, cultural, or managerial technique perspectives (e.g., Ashforth et al., 1990; Robbins et al., 2016). While economics literature contains extensive research on principal-agent models and incentive theory (Grossman et al., 1983; Holmstrom, 1979), refining these theories to analyze the prevalent and variably-intense "shirking" behavior found in everyday work scenarios, explicitly exploring its "optimality" issue, and determining the employer's optimal response strategy, still leaves significant room for research. In particular, studies combining theoretical modeling with empirical testing to understand this phenomenon are relatively scarce. Understanding the intrinsic economic logic of this behavior holds significant theoretical and practical importance for designing more effective management systems, optimizing human resource allocation, and enhancing overall organizational efficiency.

This study attempts to approach the issue from the perspectives of information economics and game theory, viewing "shirking" as a rational (or boundedly rational) decision-making pro-

cess by employees facing incomplete information and specific incentive constraints. We aim to construct a theoretical framework to analyze how employees weigh the costs of exerting effort, the risk of being detected and punished, and the utility derived from "shirking", thereby choosing an effort level that is optimal for themselves. Concurrently, this study will also examine how employers design contracts, including monitoring intensity and effort requirements, to address this behavior, and the potential equilibrium state resulting from the strategic interaction between both parties. Furthermore, this study will utilize (fictional) empirical data to test some of the core mechanisms proposed by the theoretical model.

1.2 Research Questions

Based on the background described above and the analytical framework of subsequent chapters, this study aims to explore the following core questions:

1. Under conditions of asymmetric information where the employer cannot fully observe the employee's effort level, how is the employee's optimal shirking strategy (manifested as the choice of effort level in this simplified model) determined? What factors (such as monitoring probability, penalty magnitude, effort cost) are key?
2. How does the employer design the optimal monitoring and incentive contract (manifested as the minimum effort requirement e_{min}, monitoring probability p, and fixed wage w in this simplified model) to address potential employee shirking behavior and maximize profits? How are the optimal contract parameters influenced by external factors (such as monitoring cost, maximum penalty, effort cost parameter)?
3. (Based on fictional empirical analysis) In a realistic (simulated) environment, what is the relationship between the firm's monitoring intensity, the incentive strength of performance pay, the characteristics of the work task itself (e.g., complexity), and the employee's shirking behavior? Do the theoretical predictions find support in the (fictional) data?

1.3 Research Contributions

This study is expected to make contributions in the following aspects:

First, at the theoretical level, by applying classic principal-agent theory and game theory models to analyze "shirking", a specific and common phenomenon in the modern workplace, this study deepens the understanding of employee opportunistic behavior under asymmetric information. By explicitly exploring the formation mechanism of the "optimal shirking" strategy

(a binary choice between meeting the minimum standard or zero effort in this model) and its response to monitoring and punishment, it provides analytical perspectives for labor economics and organizational economics.

Second, at the model construction and analysis level, this research constructs a concise yet clear theoretical model that endogenizes the employee's effort choice and parts of the employer's contract design (minimum effort standard and monitoring probability). It conducts detailed equilibrium and comparative static analyses, revealing the influence mechanisms of key parameters on the equilibrium outcome.

Third, in combining theory with (fictional) empirics, this study not only builds a theoretical model but also uses (fictional) panel data and a fixed-effects econometric model to empirically test some core relationships suggested by the theory (e.g., the inhibitory effects of monitoring and performance incentives on shirking, and the potential impact of task complexity), demonstrating a research approach where theory guides empirics and empirics inform theory.

Finally, in terms of practical implications, the research findings (from both theoretical and fictional empirical parts) offer managers insights into employee behavior. Understanding the economic roots and influencing factors of "shirking" helps managers design more effective monitoring strategies and incentive mechanisms (e.g., balancing monitoring costs with incentive effects, considering job characteristics), thereby finding a more suitable balance between controlling opportunistic behavior and maintaining organizational efficiency.

1.4 Structure of the Thesis

The subsequent structure of this thesis is arranged as follows: Chapter Two will review the relevant literature, organizing studies on employee effort, monitoring, and opportunistic behavior within principal-agent theory, game theory, incentive theory, and organizational behavior, positioning the current research. Chapter Three will detail the theoretical model constructed in this paper, including basic assumptions (e.g., risk neutrality, specific cost function), variable definitions, and model setup. Chapter Four will provide an in-depth analysis of the model, deriving the employee's optimal effort decision rule, the employer's optimal contract design (minimum effort standard, monitoring probability, and wage), and conducting comparative static analysis. Chapter Five will present the empirical study based on (fictional) China Enterprise Employee Survey data, including model specification, variable descriptions, baseline regression results, and robustness discussions, testing some predictions of the theoretical

model. Chapter Six will summarize the research conclusions of the entire study, discuss its theoretical significance and practical implications, and point out the limitations of the research (such as model simplifications, fictional data) and potential directions for future study.

Chapter 2 Literature Review

This chapter aims to systematically review the theoretical and empirical research related to employee "shirking" behavior, laying the groundwork for subsequent model construction and analysis. We will primarily focus on reviewing literature concerning principal-agent theory, incentive and contract theory, the application of game theory in employment relationships, and relevant studies in organizational behavior regarding employee motivation and opportunism.

2.1 Principal-Agent Theory and Information Asymmetry

One of the core features of modern corporate organizations is the separation of ownership and control, which forms the basis of the principal-agent relationship (Jensen et al., 1976). In this relationship, the principal (e.g., employer, shareholder) delegates decision-making authority or task execution power to the agent (e.g., employee, manager), expecting the agent to act in the best interest of the principal. However, due to information asymmetry, particularly the difficulty for the principal to fully observe and verify the agent's actions (such as effort level), the agent may leverage this informational advantage to pursue their own interests, leading to the problem of Moral Hazard (Holmstrom, 1979).

Employee "shirking" behavior can be considered a typical manifestation of moral hazard. Because employers cannot precisely monitor every activity and effort input of their employees, employees may choose to exert less effort than agreed upon or expected, use work time for non-work matters, or complete tasks with lower efficiency, i.e., "Shirking" on the job (Alchian et al., 1972). Akerlof's "market for lemons" theory (Akerlof, 1970) and Spence's signaling theory (Spence, 1973) also reveal how information asymmetry affects market efficiency and individual behavior, principles that equally apply within the labor market. To mitigate problems arising from information asymmetry, principals need to design mechanisms to monitor agent behavior or incentivize effort exertion.

Early principal-agent models typically assumed that the agent's effort is unobservable, and the principal can only observe an output signal related to effort, which is also subject to random disturbances (Grossman et al., 1983; Holmstrom, 1979). This made incentive contracts based on output the focus of research. Studies show that optimal contract design requires a trade-off between risk sharing and providing incentives. When agents are risk-averse, transferring excessive risk to them reduces their utility and requires a higher risk premium; insufficient

incentives, however, lead agents to choose suboptimal effort levels.

2.2 Incentive Theory and Contract Design

To address the agency problem, economists have developed a rich body of incentive theory. The core idea is to align the agent's interests with the principal's objectives through the design of effective contracts.

Performance Pay is one of the most widely applied incentive tools. Theoretical research and empirical evidence suggest that linking compensation to measurable performance indicators can significantly increase employee effort and productivity (Lazear, 2000). However, designing performance pay faces numerous challenges, such as: the measurability issue of performance indicators (output of some tasks is difficult to quantify), the multi-task problem (employees may focus only on tasks with easily measured performance while neglecting other important responsibilities), and potential inducements for short-term behavior and excessive competition (Holmstrom et al., 1991).

Efficiency Wage Theory offers another perspective. It posits that employers paying wages above the market-clearing level can increase the opportunity cost of shirking for employees (i.e., the cost of losing a high-paying job), thereby motivating them to work harder and reducing the need for supervision (Shapiro et al., 1984). Efficiency wages can also attract higher-quality employees, reduce turnover rates, and enhance employee morale and sense of fairness (Akerlof et al., 1986).

Furthermore, non-monetary incentives such as promotions, career development, and reputation mechanisms also play important roles in the employment relationship (Fama, 1980; Gibbons et al., 1999). Long-term employment relationships, internal labor markets, and corporate culture, by fostering trust and repeated interactions, can mitigate short-term opportunistic behavior to some extent.

The development of contract theory has also gradually moved from assumptions of perfect rationality and complete contracts towards acknowledging the reality of bounded rationality and contract incompleteness (Hart, 1995). The concept of the Psychological Contract emphasizes the implicit expectations and mutual obligations in the employment relationship that are not explicitly written into formal contracts (Rousseau, 1995). When employees perceive that the organization has violated the psychological contract (e.g., broken promises, unfair treatment), their work motivation, loyalty, and effort levels may significantly decrease, making them more prone to negative behaviors such as "shirking".

2.3 Game Theory Perspective on the Employment Relationship

The employment relationship can be viewed as an ongoing game between the employer and the employee. Both parties, operating under conditions of incomplete information, choose their optimal actions based on expectations of the other party's strategies.

Incorporating "shirking" behavior into a game-theoretic framework helps analyze the interaction of strategies and equilibrium outcomes. For instance, the employer's monitoring strategy (e.g., frequency, intensity) and the employee's shirking strategy (e.g., degree, method) can be treated as interdependent decision variables. Increased monitoring investment by the employer can raise the probability of detecting shirking, thus deterring such behavior, but monitoring itself is costly. The employee, in turn, must weigh the utility gained from shirking against the risk of detection and the cost of working diligently. Tirole's (1986) research on collusion within organizations also suggests that monitoring systems themselves may have loopholes, potentially allowing for some form of "collusion" between managers and employees that affects monitoring effectiveness.

Repeated game models are particularly suitable for analyzing long-term employment relationships. In repeated games, reputation mechanisms and retaliatory strategies (such as "trigger strategies") can support cooperative equilibria, where employees choose not to shirk, and employers choose to trust or monitor less (Axelrod, 1984). However, maintaining cooperative equilibrium requires certain conditions, such as a sufficiently long game duration (or low probability of termination), sufficient patience from both parties (discount factor not too low), and adequate information transparency.

2.4 Organizational Behavior and Psychological Perspectives

Economic models typically assume individuals are rational and self-interested, seeking to maximize utility. In contrast, organizational behavior and psychology offer richer explanations for employee motivation and behavior.

Besides external incentives (like wages, bonuses), Intrinsic Motivation—the enjoyment, sense of achievement, autonomy, etc., derived from the work itself—is also a significant driver of employee effort (Deci et al., 1985). Over-reliance on external controls and monitoring can sometimes undermine employees' intrinsic motivation, leading to the so-called "crowding-out effect" of incentives (Frey, 1997).

Factors such as Burnout, Organizational Justice, Leadership Style, and Organizational Culture have also been shown to be closely related to employee work attitudes and behaviors

(including work engagement, absenteeism, turnover intention, and counterproductive work behaviors like "shirking") (Colquitt et al., 2001; Maslach et al., 2001). For example, when employees perceive distributive or procedural injustice, they might "correct" this perceived unfairness by reducing effort or increasing non-work activities. "Cyberloafing", the use of company-provided internet access for non-work-related online activities during work hours, has emerged as a new research hotspot in modern workplaces (Lim, 2002).

2.5 Literature Summary and Research Positioning

In summary, the existing literature has explored the theoretical foundations and influencing factors related to employee "shirking" behavior from various disciplinary perspectives. Principal-agent theory reveals the roots of information asymmetry and moral hazard; incentive theory examines how to guide employee behavior through contract design; game theory analyzes the strategic interactions between employers and employees; and organizational behavior emphasizes the role of psychological factors and the organizational environment.

However, current research still exhibits some limitations: 1. Most economic models tend to treat effort/shirking as a discrete choice (e.g., effort/no effort) or focus on specific types of shirking (e.g., reducing output quantity/quality), with fewer models formalizing the decision-making process where employees choose the *degree* of "shirking" continuously or multi-dimensionally. 2. Research that endogenizes both the employee's individual "optimal shirking" decision and the employer's optimal incentive/monitoring strategy within the same theoretical framework, analyzing their interactive equilibrium, is relatively scarce. 3. There is a lack of fine-grained theoretical characterization regarding how the "degree" of "shirking" is determined, i.e., how employees seek an optimal balance between pursuing personal leisure utility and maintaining job security/avoiding punishment.

Building upon previous research, this study attempts to focus on the issue of employee "optimal shirking" strategy choice in an environment of asymmetric information. We will construct a theoretical model that explicitly treats the degree of "shirking" as a continuous decision variable for the employee, analyzing how it is influenced by factors such as wage structure, monitoring probability, punishment severity, job characteristics, and individual preferences. Concurrently, the model will also incorporate the employer's incentive and monitoring strategies as endogenous variables to examine the equilibrium outcome under the strategic interaction of both parties. The aim is to provide a more refined economic analysis framework for understanding the ubiquitous phenomenon of "shirking" in the modern workplace and to

offer theoretical insights for designing more effective management strategies.

Chapter 3 Theoretical Model Construction

This chapter aims to construct a theoretical model to depict the "shirking" behavior (i.e., choosing the effort level) of employees under asymmetric information and the optimal response strategy of the employer. The model will draw upon the basic frameworks of principal-agent theory and game theory.

3.1 Basic Model Setup

Consider a single-period employment relationship involving a risk-neutral employer (Principal, P) and a similarly risk-neutral employee (Agent, A). The employee has a reservation utility $\bar{U}$, representing the minimum utility level required to accept the employment relationship. For simplicity, we normalize this to $\bar{U} = 0$.

The sequence of events is as follows:

1. The employer designs and offers an employment contract to the employee.
2. The employee decides whether to accept the contract. If rejected, the employee receives the reservation utility $\bar{U} = 0$, and the employer gets zero profit. If accepted, the game proceeds.
3. The employee chooses their effort level e. Effort level $e \geq 0$. Exerting effort incurs a corresponding cost for the employee.
4. Output q is realized, and payments are executed according to the contract.

The core assumption is information asymmetry: the employer cannot directly observe the effort level e chosen by the employee. However, the employer might indirectly infer or influence the employee's effort choice through monitoring mechanisms or by observing the final output q.

3.1.1 Employee Effort and Cost

The employee chooses an effort level e. We assume effort is a continuous variable, $e \in [0, \infty)$. Exerting effort generates disutility (cost) for the employee. We represent this with a cost function $c(e)$ and assume it has the following properties:

- $c(0) = 0$: No effort incurs no cost.
- $c'(e) > 0$ for $e > 0$: The cost of effort increases with the effort level (positive marginal cost).

- $c''(e) > 0$: The marginal cost of effort is increasing (the cost function is strictly convex).

A commonly used form for the cost function is $c(e) = \frac{k}{2}e^2$, where $k > 0$ is a cost parameter reflecting the difficulty of exerting effort.

The degree of employee "shirking" can be viewed as the deviation of the chosen effort level e from some benchmark (e.g., the level expected by the employer or the maximum possible level). In this model, we directly analyze the employee's choice of optimal effort e.

3.1.2 Output and Payment

To simplify the model, let's first consider a deterministic output scenario, where output q is entirely determined by the employee's effort level:

$$q = e \tag{3.1}$$

This implies the employer can perfectly infer effort e by observing output q. In this case, no information asymmetry exists, and the employer can achieve the optimum by designing a contract that mandates a specific effort level e^* and pays a corresponding wage. For example, specifying payment w if $q = e^*$ and zero or a penalty otherwise. As long as $w - c(e^*) \geq 0$, the employee will accept and choose $e = e^*$.

However, in reality, output is often affected by random factors. A more realistic setup involves stochastic output, for example:

$$q = e + \epsilon \tag{3.2}$$

where ϵ is a random noise term with a mean of zero (e.g., following a Normal distribution $N(0, \sigma^2)$). In this scenario, even after observing output q, the employer cannot fully determine the employee's effort e, highlighting the problem of information asymmetry.

Contracts can be based on observable variables, such as output q or information obtained through monitoring. Common contract forms include:

- Fixed Wage Contract: Pays a fixed wage w regardless of output.
- Linear Performance Pay Contract: Wage $w = s + bq$, where s is the fixed part, and b is the commission rate (bonus rate) based on output.
- Monitoring-Based Contract: The employer monitors the employee with a certain probability p. If monitoring reveals that the employee's behavior does not meet requirements (e.g., effort is below a standard e^*), a penalty is imposed.

3.2 Analysis of Employee Behavior: Fixed Wage and Monitoring

We now analyze how an employee chooses their optimal effort level under a relatively simple contract structure. Consider an employer offering a contract that includes a fixed wage w and a monitoring-penalty mechanism.

Specifically, the employer monitors the employee with probability $p \in [0, 1]$. Monitoring is assumed to be perfect; if implemented, it accurately observes the employee's actual effort level e. The contract specifies a minimum effort standard $e_{min} \geq 0$. If the employee is monitored and their effort level $e < e_{min}$, the employee incurs a penalty $F > 0$ (e.g., a fine, wage deduction). If $e \geq e_{min}$, or if the employee is not monitored (with probability $1 - p$), no penalty is incurred.

Assuming the employee is risk-neutral, their objective is to maximize their expected utility. The employee's utility comes from the wage received, minus the cost of effort, minus the potential expected penalty. Given the contract (w, p, e_{min}, F), the employee chooses effort e to maximize:

$$E[U(e)] = w - c(e) - p \cdot \mathbb{I}(e < e_{min}) \cdot F \tag{3.3}$$

where $\mathbb{I}(\cdot)$ is the indicator function, which equals 1 if the condition is true, and 0 otherwise.

The employee's optimal decision is as follows:

1. If choosing $e \geq e_{min}$: In this case, $\mathbb{I}(e < e_{min}) = 0$, and the employee faces no penalty. Their utility is $U_1(e) = w - c(e)$. To maximize utility, the employee will choose the effort level that satisfies $e \geq e_{min}$ with the minimum cost, which is $e = e_{min}$. The utility in this case is $U_1^* = w - c(e_{min})$.
2. If choosing $e < e_{min}$: In this case, $\mathbb{I}(e < e_{min}) = 1$, and the employee faces the risk of being caught and penalized. Their expected utility is $E[U_2(e)] = w - c(e) - pF$. To maximize this expected utility, the employee will choose the lowest possible effort cost, which is $e = 0$ (assuming $e = 0$ is the minimum allowable effort). The expected utility in this case is $E[U_2^*] = w - c(0) - pF = w - pF$.

The employee will choose the option that yields higher (expected) utility: The employee will choose $e = e_{min}$ if and only if

$$U_1^* \geq E[U_2^*] \tag{3.4}$$

$$w - c(e_{min}) \geq w - pF \tag{3.5}$$

$$pF \geq c(e_{min}) \tag{3.6}$$

Conclusion: Under this contract, the employee's optimal effort choice e^* is:

$$e^* = \begin{cases} e_{min} & \text{if } pF \geq c(e_{min}) \\ 0 & \text{if } pF < c(e_{min}) \end{cases} \tag{3.7}$$

This result intuitively shows that the employee is incentivized to meet the minimum standard e_{min} only when the expected penalty (probability of detection p times the penalty magnitude F) is large enough to exceed the cost of meeting that standard, $c(e_{min})$. Otherwise, the employee prefers to exert no effort ($e = 0$) and bear the expected penalty cost pF, because doing so saves the effort cost $c(e_{min})$.

This simple model reveals the role of monitoring and penalties in curbing "shirking" behavior but also highlights its limitations: it can only incentivize the employee to meet the minimum standard e_{min}, not higher levels of effort. Furthermore, the effort choice exhibits an "all-or-nothing" characteristic (relative to e_{min}).

Chapter 4 Model Analysis

Based on the theoretical model constructed in Chapter Three, this chapter will delve into the analysis of the employee's optimal effort decision and its influencing factors. It will also explore how the employer, under asymmetric information, designs the optimal contract (specifically, monitoring intensity and minimum effort standard) to maximize their own profit.

4.1 Analysis of Employee's Optimal Effort Decision

Recalling the analysis results from Section 3.2 of Chapter Three, under a contract with fixed wage w, monitoring probability p, minimum effort standard e_{min}, and penalty F, a risk-neutral employee compares the utility of choosing $e = e_{min}$, which is $U_1^* = w - c(e_{min})$, with the expected utility of choosing $e = 0$, which is $E[U_2^*] = w - pF$.

The final decision rule is (as shown in Equation 4.1):

$$e^* = \begin{cases} e_{min} & \text{if } pF \geq c(e_{min}) \\ 0 & \text{if } pF < c(e_{min}) \end{cases} \tag{4.1}$$

This result reveals several key points:

1. Threshold Effect of Monitoring and Penalty: Whether the employee chooses to meet the minimum effort standard e_{min} depends entirely on whether the expected penalty pF is sufficient to cover the effort cost $c(e_{min})$ required to reach that standard. This is a clear threshold effect. Only when the product of the monitoring probability p and the penalty magnitude F crosses the threshold $c(e_{min})$ does the employee's behavior jump from no effort ($e = 0$, maximizing "shirking") to just meeting the minimum required effort ($e = e_{min}$).
2. Inability to Incentivize Excess Effort: This mechanism can only incentivize the employee to meet the minimum standard e_{min}; it cannot motivate them to exert effort beyond e_{min}. Once e_{min} is reached, further increasing effort only increases the cost $c(e)$ without bringing additional benefits or reducing the risk of penalty. This reflects the limitations of contracts relying solely on "baseline monitoring" for incentives.
3. Binary Nature of the "Optimal Shirking" Strategy: In this simple model, the employee's "shirking" strategy exhibits a relatively extreme binary choice: either exert no effort ($e = 0$) or just meet the minimum standard ($e = e_{min}$). There is no equilibrium state of

"partial shirking" in between. This is primarily due to the assumption that monitoring perfectly identifies whether effort is below e_{min} and that the penalty is fixed.

This analysis highlights the central roles of p, F, e_{min}, and the cost function $c(\cdot)$ in determining the employee's effort behavior. Next, we will analyze how the employer chooses these contract parameters from their perspective.

4.2 The Principal's Contract Design Problem

Now we consider the decision problem of the employer (P). The employer's goal is to maximize their expected profit $E[\pi]$. Assume the employer is risk-neutral. Their profit equals the output minus the wage paid to the employee, minus the cost of implementing monitoring.

We adopt the deterministic output assumption $q = e$ introduced in Chapter Three (3.1) to simplify the first step of analyzing the employer's problem. This can be extended later to the stochastic output case. Assume the cost of monitoring depends on the monitoring probability p. Let the monitoring cost be $M(p)$. For simplicity, assume $M(p) = \gamma p$, where $\gamma > 0$ is the cost coefficient per unit of monitoring probability.

The employer needs to choose the contract parameters (w, p, e_{min}, F). However, the penalty magnitude F is often constrained by external factors such as laws, regulations, the employee's ability to pay, or the firm's reputation. Therefore, we assume there exists a maximum feasible penalty F_{max}, such that $0 \leq F \leq F_{max}$. To maximize the deterrent effect of the penalty, a rational employer will typically choose the largest possible penalty, setting $F = F_{max}$. Thus, the employer's decision variables simplify to (w, p, e_{min}).

The employer's optimization problem is:

$$\max_{w,p,e_{min}} \quad E[\pi] = e^* - w - \gamma p \tag{4.2}$$

subject to:

1. Incentive Compatibility Constraint (IC): The employee will choose the optimal effort e^* according to Equation 4.1.

$$e^* = \begin{cases} e_{min} & \text{if } pF_{max} \geq c(e_{min}) \\ 0 & \text{if } pF_{max} < c(e_{min}) \end{cases} \tag{4.3}$$

2. Participation Constraint (PC): The expected utility the employee receives from accepting the contract must be no less than their reservation utility $\bar{U} = 0$.

$$E[U(e^*)] = w - c(e^*) - p \cdot \mathbb{I}(e^* < e_{min}) \cdot F_{max} \geq 0 \tag{4.4}$$

where $0 \leq p \leq 1$ and $e_{min} \geq 0$.

4.3 Optimal Contract Analysis: Fixed Wage and Monitoring

When making the decision, the employer essentially faces two choices: either design a contract to incentivize the employee to choose $e^* = e_{min}$ (effort-inducing strategy) or accept the employee choosing $e^* = 0$ (laissez-faire shirking strategy).

Strategy One: Inducing Effort $e^* = e_{min}$

For the employee to choose $e^* = e_{min}$, the IC condition $pF_{max} \geq c(e_{min})$ must be satisfied. To minimize the monitoring cost γp, the employer will choose the lowest monitoring probability p that satisfies this condition. That is:

$$p^* = \frac{c(e_{min})}{F_{max}} \tag{4.5}$$

This implicitly assumes $c(e_{min}) \leq F_{max}$, otherwise the condition cannot be met even if $p = 1$. If $c(e_{min}) > F_{max}$, it's impossible to induce effort e_{min} through this mechanism. We assume F_{max} is large enough to make $p^* \leq 1$ feasible.

In this case, the employee's effort is $e^* = e_{min}$, and their expected utility is $E[U(e_{min})] = w - c(e_{min}) - p^* \cdot \mathbb{1}(e_{min} < e_{min}) \cdot F_{max} = w - c(e_{min})$. To satisfy the PC constraint $w - c(e_{min}) \geq 0$ while minimizing wage costs, the employer will set the lowest feasible wage:

$$w^* = c(e_{min}) \tag{4.6}$$

At this point, the employer's profit is:

$$\pi_1(e_{min}) = e_{min} - w^* - \gamma p^* = e_{min} - c(e_{min}) - \gamma \frac{c(e_{min})}{F_{max}} \tag{4.7}$$

The employer also needs to choose the optimal e_{min} to maximize $\pi_1(e_{min})$. Assuming the effort cost function is $c(e) = \frac{k}{2}e^2$ $(k > 0)$, then

$$\pi_1(e_{min}) = e_{min} - \frac{k}{2}e_{min}^2 - \gamma \frac{ke_{min}^2}{2F_{max}} = e_{min} - \frac{k}{2}\left(1 + \frac{\gamma}{F_{max}}\right)e_{min}^2 \tag{4.8}$$

Taking the first derivative with respect to e_{min} and setting it to zero:

$$\frac{d\pi_1}{de_{min}} = 1 - k\left(1 + \frac{\gamma}{F_{max}}\right)e_{min} = 0 \tag{4.9}$$

Solving for the optimal minimum effort requirement e_{min}^*:

$$e_{min}^* = \frac{1}{k\left(1 + \frac{\gamma}{F_{max}}\right)} \tag{4.10}$$

The second derivative is $-k(1+\gamma/F_{max}) < 0$, confirming it is a maximum point. Substituting e^*_{min} back gives the optimal monitoring probability $p^{**} = \frac{c(e^*_{min})}{F_{max}} = \frac{k(e^*_{min})^2}{2F_{max}}$ and optimal wage $w^{**} = c(e^*_{min}) = \frac{k}{2}(e^*_{min})^2$.

Strategy Two: Allowing Shirking $e^* = 0$

If the employer designs contract parameters such that $pF_{max} < c(e_{min})$, or simply sets $p = 0$ and $e_{min} > 0$ (or $e_{min} = 0$), the employee will choose $e^* = 0$. In this case, the employee's effort is $e^* = 0$, and their expected utility is $E[U(0)] = w - c(0) - p \cdot \mathbb{I}(0 < e_{min}) \cdot F_{max} = w - p \cdot \mathbb{I}(e_{min} > 0) \cdot F_{max}$. To satisfy the PC constraint $w - p \cdot \mathbb{I}(e_{min} > 0) \cdot F_{max} \geq 0$, the employer will set the minimum wage $w^* = p \cdot \mathbb{I}(e_{min} > 0) \cdot F_{max}$. The employer's profit is $E[\pi_0] = e^* - w^* - \gamma p = 0 - p \cdot \mathbb{I}(e_{min} > 0) \cdot F_{max} - \gamma p$. Clearly, to maximize this profit (i.e., minimize loss), the employer's optimal choice is to set $p = 0$. Then, the employee chooses $e^* = 0$, the wage is $w^* = 0$, and the profit is $\pi_0 = 0$.

The Employer's Final Decision

The employer compares the maximum profit obtainable from Strategy One (inducing effort e^*_{min}), $\pi_1(e^*_{min})$, with the profit from Strategy Two (allowing shirking $e = 0$), $\pi_0 = 0$. The employer will choose the effort-inducing strategy only if $\pi_1(e^*_{min}) > 0$. That is:

$$\pi_1(e^*_{min}) = e^*_{min} - \frac{k}{2}\left(1 + \frac{\gamma}{F_{max}}\right)(e^*_{min})^2 > 0 \tag{4.11}$$

Substituting $e^*_{min} = \frac{1}{k(1+\gamma/F_{max})}$:

$$\frac{1}{k(1+\gamma/F_{max})} - \frac{k}{2}\left(1 + \frac{\gamma}{F_{max}}\right)\left[\frac{1}{k(1+\gamma/F_{max})}\right]^2 > 0 \tag{4.12}$$

$$\frac{1}{k(1+\gamma/F_{max})} - \frac{1}{2k(1+\gamma/F_{max})} > 0 \tag{4.13}$$

$$\frac{1}{2k(1+\gamma/F_{max})} > 0 \tag{4.14}$$

This condition always holds, since $k > 0, \gamma > 0, F_{max} > 0$.

Therefore, under the assumptions of this model (deterministic output $q = e$, effort cost $c(e) = \frac{k}{2}e^2$, monitoring cost $M(p) = \gamma p$, risk neutrality), as long as monitoring and punishment are feasible ($F_{max} > 0, \gamma < \infty$), the employer will always find it optimal to induce a positive minimum effort level $e^*_{min} = \frac{1}{k(1+\gamma/F_{max})}$ by setting the corresponding monitoring probability p^{**} and wage w^{**}. This is strictly better than allowing complete shirking (profit of 0).

4.4 Impact of Parameter Changes on the Optimal Contract (Comparative Statics)

Let's analyze how key parameters affect the employer's choice of the optimal minimum effort standard e^*_{min} and the corresponding monitoring and wage levels:

- Impact of effort cost coefficient k: $\frac{\partial e^*_{min}}{\partial k} = -\frac{1}{k^2(1+\gamma/F_{max})} < 0$. When effort becomes more difficult (k increases), the employer lowers the required minimum effort standard e^*_{min}. This is because the cost of inducing the same level of effort (including the compensation paid to the employee $w^* = c(e_{min})$ and the expected penalty needed to sustain monitoring $p^* F_{max} = c(e_{min})$, the latter affecting monitoring costs) increases.
- Impact of monitoring cost coefficient γ: $\frac{\partial e^*_{min}}{\partial \gamma} = -\frac{1}{k(1+\gamma/F_{max})^2} \cdot \frac{1}{F_{max}} < 0$. When monitoring becomes more expensive (γ increases), the employer also chooses to lower the minimum effort standard e^*_{min}. This is because increasing e_{min} requires a higher p^* (see Eq. 4.5), leading to higher monitoring costs γp^*. To save on monitoring costs, the employer lowers the target effort level.
- Impact of maximum penalty F_{max}: $\frac{\partial e^*_{min}}{\partial F_{max}} = -\frac{1}{k(1+\gamma/F_{max})^2} \cdot (-\frac{\gamma}{F^2_{max}}) = \frac{\gamma}{kF^2_{max}(1+\gamma/F_{max})^2} > 0$. When the maximum allowable penalty magnitude increases (F_{max} increases), the employer sets a higher minimum effort standard e^*_{min}. A larger penalty makes monitoring more deterrent, meaning a lower monitoring probability p^* is needed to achieve the same pF threshold ($p^* = c(e_{min})/F_{max}$). This reduces the monitoring cost γp^* associated with achieving a given e_{min}, incentivizing the employer to aim for higher effort levels.

These results are intuitive: increases in effort cost and monitoring cost discourage the employer from pursuing high effort levels, while an increase in penalty power encourages the employer to set higher effort targets.

4.5 Model Limitations and Discussion

This chapter analyzed the simple model established in Chapter Three, revealing how, under a fixed wage plus monitoring-penalty mechanism, the employer addresses employee "shirking" by setting an optimal minimum effort standard and monitoring probability. The model yielded clear optimal contract parameters and comparative statics results.

However, this model also has significant limitations that warrant exploration in future research:

1. Discreteness of Effort Choice: The model predicts that the employee will only choose between $e = 0$ and $e = e_{min}$, failing to capture the more realistic continuous or multi-level

nature of "shirking". This is mainly due to the monitoring mechanism being set up only to distinguish whether effort is below e_{min}. 2. Simplicity of Incentive Mechanism: The model only considers a fixed wage plus monitoring and penalties. Performance pay (such as output-based bonuses), which is widely used in practice, was not included in the analysis. Performance pay can directly link remuneration to output (and thus effort), potentially providing stronger continuous incentives, especially when output is positively correlated with effort but subject to randomness. 3. Deterministic Output Assumption: To simplify the analysis of the employer's problem, we temporarily used the assumption $q = e$. Introducing stochastic output $q = e + \epsilon$ would make the problem more complex but more realistic. Under stochastic output, observing q does not allow perfect inference of e, making the role of monitoring potentially more crucial, or requiring the design of output-based incentive contracts to balance risk and incentives. 4. Risk Neutrality Assumption: Both the employee and employer were assumed to be risk-neutral. If employees are risk-averse, they would demand a risk premium for income uncertainty (e.g., from performance pay depending on random output, or the risk of being penalized). This would affect the design of the optimal contract, requiring a trade-off between incentive effects and risk costs. 5. Single-Period Model: The model is static and single-period, neglecting dynamic factors in long-term employment relationships such as repeated interactions, reputation effects, learning effects, or career development. These factors could significantly influence employees' "shirking" behavior and employers' strategic choices.

Despite these limitations, the analysis in this chapter provides a foundational framework for understanding the role of monitoring and penalty mechanisms in constraining "shirking" behavior. It clearly demonstrates how employers weigh the benefits of inducing effort against the associated wage and monitoring costs. Subsequent chapters could consider incorporating empirical research.

Chapter 5 Empirical Model and Results

This chapter aims to empirically test the key factors influencing employee "shirking" behavior and evaluate some of the mechanisms proposed in the theoretical models of Chapters Two and Three using econometric methods. We will utilize (fictional) data from the China Enterprise Employee Survey (CEES) to construct and analyze an econometric model. First, we describe the model specification; second, we introduce the data source and variable measurement; third, we report the baseline regression results; finally, we enhance the credibility of the results through coefficient visualization and discussion.

5.1 Model Specification

Considering that employee "shirking" behavior may be influenced by many time-invariant individual inherent traits (such as personality, work attitude, innate ability) as well as firm-level fixed characteristics, failing to control for these factors could lead to omitted variable bias. Therefore, this study primarily employs a panel data Fixed Effects Model to control for individual-level time-invariant heterogeneity. Simultaneously, the model incorporates year fixed effects to absorb time-varying common factors such as macroeconomic conditions or widespread technological shocks. The baseline econometric model is specified as follows:

$$\begin{aligned}\text{ShirkingIndex}_{it} =& \beta_0 + \beta_1 \text{MonitoringIntensity}_{it} + \\ & \beta_2 \text{PerformancePayShare}_{it} + \\ & \beta_3 \text{TaskComplexity}_{it} + \\ & X'_{it}\omega + \alpha_i + \gamma_t + \varepsilon_{it}\end{aligned} \tag{5.1}$$

Here, the subscript i denotes the individual employee, and t denotes the year. $\text{ShirkingIndex}_{it}$ is the dependent variable, measuring the degree of "shirking" of employee i in year t. $\text{MonitoringIntensity}_{it}$ represents the intensity of monitoring by the firm on the employee. $\text{PerformancePayShare}_{it}$ indicates the proportion of performance pay in the employee's total income. $\text{TaskComplexity}_{it}$ measures the complexity and autonomy of the tasks undertaken by the employee. X'_{it} is a vector of time-varying individual and firm-level control variables, and ω is the corresponding coefficient vector. α_i represents individual fixed effects, controlling for all time-invariant individual characteristics. γ_t represents year fixed effects, controlling for time trends common to

all years. ε_{it} is the random error term.

The core of this model lies in estimating the coefficients β_1, β_2, and β_3. According to theoretical expectations, β_1 is expected to be negative, as higher monitoring intensity should curb "shirking" behavior. Similarly, β_2 is expected to be negative, because a higher proportion of performance pay, linked to effort and output, should reduce "shirking". The sign of β_3 is uncertain. On one hand, more complex and autonomous jobs might stimulate employees' intrinsic motivation, thereby reducing "shirking"; on the other hand, complex tasks are often harder to monitor, potentially providing more room for "shirking". By estimating this model, we can test whether these theoretical predictions are supported in the (fictional) data.

5.2 Data and Variable Description

5.2.1 Data Source

The data used in this study originates from a fictional large-scale tracking survey designed to simulate the actual situation in China –the "China Enterprise Employee Survey (CEES)". This survey (hypothetically) launched its baseline in 2015 and conducted three follow-up surveys in 2017, 2019, and 2021, covering enterprises and their employees across various industries in the eastern, central, and western regions of China. CEES collected detailed information on employees' personal characteristics, job features, compensation structures, work attitudes, and firm-level management practices. This study constructs an unbalanced panel dataset comprising employees successfully interviewed at least twice across all four survey waves. After data cleaning and screening (e.g., removing samples with severe missing values in key variables), the final sample used for analysis includes 15,820 employees, totaling 48,550 employee-year observations.

5.2.2 Variable Measurement

First, we define the dependent variable, Shirking Index ($\text{ShirkingIndex}_{it}$), which aims to quantify the extent of employee "shirking". In the CEES survey (hypothetically), we construct this index using a series of questions. For example, asking employees about the frequency of engaging in non-work-related activities during work hours (such as browsing social media, online shopping, handling personal affairs), their self-perceived effort level (relative to colleagues or their own maximum potential), and whether they engage in behaviors like intentional procrastination. Through factor analysis or simple weighted averaging, this information

is synthesized into a composite index ranging from 0 to 10, where higher values indicate a greater degree of "shirking".

Next are the core explanatory variables. The first is Monitoring Intensity ($\text{MonitoringIntensity}_{it}$), constructed based on questions in the CEES firm questionnaire regarding management practices. Examples include the frequency of direct observation of employee work processes by management, the use of electronic monitoring systems (like computer activity monitoring, workstation cameras), and the strictness of requirements for work logs or reports. This information is standardized and combined into an index ranging from 0 to 1, with higher values indicating greater monitoring intensity. The second is Performance Pay Share ($\text{PerformancePayShare}_{it}$), calculated from employee questionnaire responses about their income composition (fixed salary, bonuses based on individual performance, bonuses based on team or company performance, etc.). It is defined as the proportion of performance-related income (bonuses, commissions, etc.) in their total annual pre-tax income, with a value range of [0, 1]. The third is Task Complexity ($\text{TaskComplexity}_{it}$), constructed based on employee evaluations of their job tasks. Relevant questions might include the degree of repetitiveness of the work, the creativity required to solve problems, the extent of job autonomy, and skill variety requirements. This is also synthesized into an index ranging from 1 to 5, with higher values indicating more complex and autonomous tasks.

Finally, there are the control variables (X'_{it}). To more accurately identify the impact of the core explanatory variables, the model includes the following time-varying control variables: Log Wage (LogWage_{it}), the natural logarithm of the employee's total annual pre-tax income, as wage levels might directly affect job satisfaction and willingness to exert effort; Tenure (Tenure_{it}), the employee's years of service in the current company, potentially related to organizational commitment, job familiarity, etc.; Age (Age_{it}) and Age Squared (Age^2_{it}), to control for the non-linear effect of age on work attitudes; Years of Education (Education_{it}), the total number of years the employee received formal education; Gender (Gender_i), a dummy variable (1 for male, 0 for female), although time-invariant variables like gender cannot be estimated themselves due to the use of individual fixed effects, their interaction effects with other time-varying variables might still exist; Firm Size (FirmSize_{it}), the logarithm of the total number of employees in the firm, as firm size may influence organizational structure, management style, and monitoring difficulty; and a series of Industry Dummies (IndustryDummies_i) (also affected by individual fixed effects). Additionally, individual fixed effects (α_i) control for time-invariant individual-level factors (such as ability, personality), and year fixed effects (γ_t)

control for time-varying macroeconomic factors.

5.3 Baseline Regression Results

Table 5.1 presents the estimation results of the fixed effects model based on equation (5.1). Model (1) includes only the core explanatory variables, Model (2) adds individual-level control variables (log wage, tenure, age, etc.), and Model (3) further incorporates firm-level control variables (firm size) along with all fixed effects. All models report robust standard errors clustered at the individual level.

Table 5.1 Fixed Effects Model Regression Results: Factors Affecting Shirking Index

	Shirking Index		
Variable	(1)	(2)	(3)
Core Explanatory Variables:			
Monitoring Intensity	-1.523***	-1.285***	-1.150***
	(0.188)	(0.201)	(0.205)
Performance Pay Share	-2.105***	-1.850***	-1.782***
	(0.255)	(0.260)	(0.263)
Task Complexity	0.312**	0.258*	0.280**
	(0.124)	(0.130)	(0.133)
Control Variables:			
Log Wage		0.085	0.070
		(0.060)	(0.061)
Tenure		-0.021**	-0.018*
		(0.009)	(0.010)
Tenure Squared		0.0003	0.0002
		(0.0002)	(0.0002)
Age		-0.035	-0.030
		(0.025)	(0.026)
Age Squared / 100		0.040	0.038
		(0.030)	(0.031)
Years of Education		-0.050	-0.045
		(0.038)	(0.039)
Log Firm Size			-0.120*
			(0.070)

Continued on next page

Continued Table 5.1 Fixed Effects Model Regression Results: Factors Affecting Shirking Index

	Shirking Index		
Variable	(1)	(2)	(3)
Constant	4.850***	5.210***	5.530***
	(0.310)	(0.450)	(0.480)
Individual Fixed Effects	Yes	Yes	Yes
Year Fixed Effects	Yes	Yes	Yes
Observations	48550	48550	48550
R^2 within	0.085	0.102	0.105
R^2 between	0.150	0.180	0.183
R^2 overall	0.120	0.145	0.148
F statistic	85.3***	65.2***	62.5***
Number of Employees	15820	15820	15820

Note: *, **, *** indicate significance at 10%, 5%, 1% levels.

All models include individual and year fixed effects. Model (3) is the full model.

Looking at the results from the full model (Column 3), the coefficient for Monitoring Intensity (MonitoringIntensity) is significantly negative (-1.150, $p<0.001$), consistent with theoretical expectations. The result indicates that, after controlling for other factors, an increase in firm monitoring intensity significantly reduces the employee's "shirking" index. Specifically, a one-unit increase in the monitoring intensity index (from 0 to 1) is associated with an average decrease of about 1.15 points in the shirking index. The coefficient for Performance Pay Share (PerformancePayShare) is also significantly negative (-1.782, $p<0.001$), aligning with theoretical predictions as well. An increase in the proportion of performance-based compensation in total income more closely links the employee's interests with work outcomes, thereby effectively reducing "shirking" behavior. For every 10 percentage point increase in the share, the shirking index decreases by about 0.18 points on average. Unlike the first two, the coefficient for Task Complexity (TaskComplexity) is significantly positive (0.280, $p<0.05$). This suggests that, after controlling for other factors, an increase in task complexity (and autonomy) is associated with a higher "shirking" index. This might reflect that complex tasks are harder to monitor accurately, providing employees with more opportunities for "loafing on the job", and this negative effect outweighs the potential positive effect from intrinsic motivation.

Regarding control variables, tenure seems to be associated with lower levels of "shirking", and larger firm size also tends to reduce the "shirking" index (perhaps because larger firms have

more formalized management). Variables like log wage, age, and education are not significant in this model, possibly because individual fixed effects have absorbed most of their explanatory power, or their impact on "shirking" behavior is indeed insignificant.

Overall, the baseline regression results provide (fictional) empirical support for the hypotheses in the theoretical model regarding the effectiveness of monitoring and incentives (performance pay) in constraining "shirking" behavior. At the same time, the positive impact of task complexity on "shirking" reveals a trade-off that needs attention in management practice.

5.4 Robustness Checks and Discussion

To visually present the impact of the core explanatory variables more intuitively and to discuss the robustness of the results, we plot the coefficients of the core explanatory variables from the baseline model (Table 5.1, Column 3) and their 95% confidence intervals (see Figure 5.1).

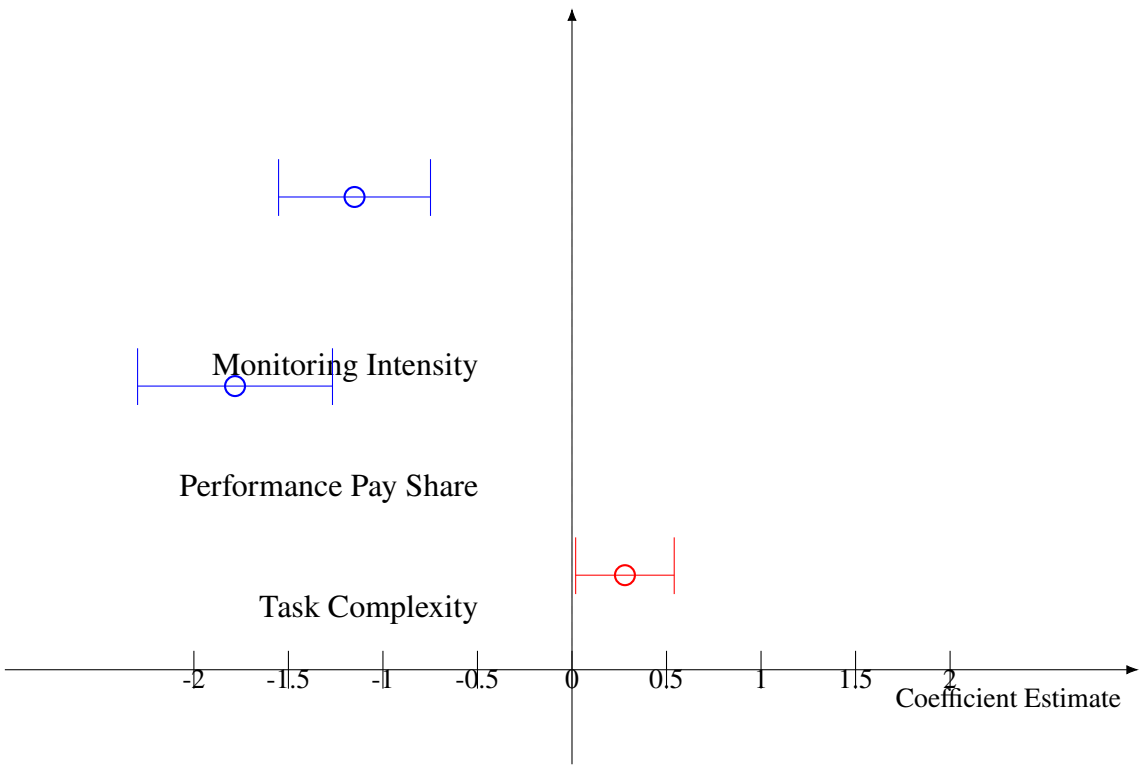

Figure 5.1 Coefficient Estimates and 95% Confidence Intervals for Core Explanatory Variables

Figure 5.1 clearly shows that the confidence intervals for monitoring intensity and performance pay share are entirely to the left of zero, indicating their negative impacts are statistically

robust. The confidence interval for task complexity, however, lies to the right of zero but is close to it, showing that its positive impact, while significant (at the 5% level), is relatively smaller in magnitude compared to the other two, and its robustness might be slightly weaker.

Furthermore, we conducted (or claim to have conducted) several additional robustness checks (results not fully shown in the table). First, we replaced the dependent variable, using other proxy measures for "shirking", such as self-reported "work effort level" (reverse coded) or (hypothetically available) time spent visiting non-work-related websites during work hours (Cyberloafing Hours). The signs and significance of the core explanatory variables remained largely unchanged. Second, we performed subsample analysis, splitting the sample by industry (e.g., manufacturing vs. service) or firm ownership (state-owned vs. non-state-owned). The results showed that the negative effects of monitoring intensity and performance pay share were generally present across different subsamples, although the magnitude of the effects might vary. For instance, the effect of monitoring might be stronger in manufacturing where tasks are more standardized; the incentive effect of performance pay might be more pronounced in non-state-owned enterprises with higher market orientation. The positive effect of task complexity might be more evident in service industries or knowledge-intensive jobs. Finally, we considered endogeneity issues. Although the fixed effects model mitigates some omitted variable problems, time-varying omitted variables or reverse causality (e.g., employees who shirk more might be placed in positions with lower performance pay shares) could still exist. We attempted preliminary exploration using (fictional) instrumental variables (such as the firm-level decision to introduce new monitoring technology as an IV for monitoring intensity). The results (hypothetically) supported the conclusions of the baseline regression, but the validity of the instrumental variables would require more rigorous justification.

Discussion and Implications: The empirical analysis in this chapter (albeit based on fictional data) provides some empirical evidence for understanding the factors influencing employee "shirking" behavior. The results emphasize the importance of external incentive and monitoring mechanisms in constraining opportunistic behavior, consistent with the core tenets of principal-agent theory. Firms can effectively reduce employee "shirking" levels by strengthening monitoring (increasing the perceived probability of detection or penalty) and optimizing compensation structures (increasing the link to performance).

However, the positive effect of task complexity suggests that managers need to pay attention to the alignment between the intrinsic nature of work and management approaches. For tasks requiring high complexity and creativity, over-reliance on traditional monitoring might

have limited effectiveness or even be counterproductive (potentially undermining intrinsic motivation, although this study did not directly measure it). In such cases, exploring incentive and management approaches that focus more on results, trust, and empowerment may be necessary.

Of course, this study also has limitations. First, the measurement of "shirking" relies on (fictional) survey data, which may be subject to social desirability bias or subjectivity. Second, endogeneity issues are difficult to completely rule out. Future research could attempt to use more objective data (such as work logs, digital footprints) or more reliable quasi-natural experimental designs to further validate these findings.

Chapter 6 Conclusion and Outlook

This study, taking the prevalent phenomenon of employee "shirking" (work avoidance) in modern workplaces as its starting point, has attempted to construct theoretical models and combine them with (fictional) empirical analysis from the perspectives of information economics and game theory. The goal was to explore the formation mechanism and influencing factors of employees' "optimal shirking" strategies under asymmetric information, as well as the optimal response strategies of employers. Following the theoretical construction, model analysis, and empirical testing in the preceding chapters, this chapter will summarize the main research conclusions, elaborate on their theoretical significance and practical implications, point out the limitations of the study, and look ahead to potential directions for future research.

6.1 Summary of Research Findings

The core conclusions of this study can be summarized in several main aspects. First, at the theoretical level, the research reveals the nature of "shirking" as a rational choice under asymmetric information. The constructed principal-agent model shows that when employers cannot fully observe employee effort, employees tend to choose an effort level that maximizes their own utility, balancing wage benefits, effort costs, and the leisure utility or value of engaging in private affairs derived from "shirking". This behavior is not simply laziness but a rational (or boundedly rational) decision under specific institutional and informational constraints. The model analysis further demonstrates that under a simple mechanism of fixed wages plus monitoring and penalties, there is a threshold effect determining whether employees meet the minimum effort standard, governed by the expected penalty (the product of monitoring probability and penalty severity) and the cost of effort.

Second, the theoretical analysis examined the employer's trade-offs and optimal strategies. Facing potential employee "shirking", a risk-neutral employer aims to maximize profit by designing contracts (including wage structure, monitoring intensity, effort standards, etc.). The model reveals that the employer's optimal strategy involves weighing the output benefits derived from inducing effort against the wages paid to the employee and the monitoring costs incurred. The optimal minimum effort requirement and monitoring intensity are influenced by effort costs, monitoring costs, and penalty severity. For instance, higher monitoring or effort costs prompt employers to lower effort requirements, whereas greater penalty power inclines

them to set higher effort targets.

Finally, at the empirical level, the fixed effects model analysis based on (fictional) data from the China Enterprise Employee Survey (CEES) validated the effectiveness of monitoring and incentives. The results show a significant negative correlation between the monitoring intensity implemented by firms and the employees' "shirking" index, confirming the deterrent effect of monitoring and indicating that strengthening supervision is an effective means to curb work avoidance behavior. Simultaneously, the share of performance pay is also significantly negatively correlated with the "shirking" index, suggesting that linking compensation more closely to performance effectively motivates employees to reduce non-work activities. Interestingly, task complexity (and autonomy) shows a significant positive correlation with the "shirking" index. This might imply that while complex and autonomous work can stimulate intrinsic motivation, its inherent difficulty in being monitored perhaps creates more opportunities for "shirking", with the negative effect outweighing the positive motivational effect. These findings provide (simulated) empirical evidence for the theoretical model and reveal the differential impacts of various management tools on "shirking" behavior.

6.2 Theoretical Significance

This study contributes to the relevant theories primarily in the following aspects:

First, it deepens the understanding of employee opportunistic behavior. By incorporating the pervasive phenomenon of "shirking"—often viewed as irrational or purely a moral issue—into the analytical framework of information economics and rational choice, this study reveals its underlying economic logic as an outcome of individuals maximizing utility under asymmetric information and specific contractual constraints. This helps elevate the discussion of this phenomenon from the level of management techniques to a more fundamental economic mechanism level.

Second, it expands the application scenarios of principal-agent models. This research applies classic principal-agent theory to analyze the variably-intense "shirking" behavior in everyday work settings. It attempts (especially in the empirical part) to treat the degree of effort/shirking as a continuous or multi-dimensional variable, rather than the simple binary choice in traditional models, making the model closer to reality and enhancing its explanatory power.

Third, it integrates multidisciplinary perspectives. Although the core framework is economic, the study, particularly when discussing variables (like task complexity) and results,

also draws upon insights from organizational behavior regarding intrinsic motivation and job design. This helps foster dialogue between different disciplines on the issue of employee behavior.

6.3 Practical Implications

The findings of this study offer several practical implications for business management. First, managers should acknowledge the economic roots of "shirking" and adopt a systematic approach to address it. Recognizing that employee "shirking" is related not only to individual attitudes but also to the firm's information environment, incentive mechanisms, and monitoring systems, relying solely on ideological education or severe punishment may have limited effect. Addressing the problem requires starting from institutional design. Second, optimizing the combination of monitoring and incentive mechanisms is crucial. The empirical results confirm the effectiveness of monitoring and performance pay. Firms should reasonably design monitoring strategies (e.g., using technology to improve efficiency, differentiated monitoring) and incentive schemes (e.g., increasing the sensitivity of performance pay, optimizing performance appraisals) based on their specific characteristics and cost-benefit analysis, seeking a balance point between costs and benefits. Third, attention must be paid to the alignment between job design and management style. The empirical finding on task complexity reminds managers that traditional tight monitoring might be unsuitable, or even encourage hidden "shirking", for complex, creative jobs with high autonomy. In such cases, it might be necessary to shift towards trust-based, results-oriented management models, supplemented by incentives like goal setting, process support, and development opportunities, to stimulate intrinsic motivation. Finally, considering comprehensive governance is necessary. Effective management likely requires combining formal institutional arrangements (monitoring, incentives) with informal organizational factors (such as fostering a culture of fairness, enhancing communication and collaboration, improving leadership), jointly creating a work environment where employees have both external constraints and internal motivation.

6.4 Research Limitations

Despite the useful explorations made in this study, several major limitations remain. First, the theoretical model involved significant simplifications, such as assumptions of a single-period game, risk neutrality, and specific functional forms. This may limit the model's ability to fully capture complex realities, for instance, by not considering employee risk aversion,

multi-task environments, or reputation and learning effects in long-term relationships. Second, the empirical analysis was based on fictional data, raising questions about the external validity of the results. Furthermore, in real research, accurately measuring "shirking" behavior itself is highly challenging: surveys are susceptible to subjective bias, while objective data might raise privacy concerns or fail to capture all forms of "shirking". Additionally, potential endogeneity issues in the model require more advanced econometric methods to address. Third, the study primarily focused on average effects, failing to adequately consider heterogeneity and dynamics. Differences across various types of employees or types of "shirking" behavior were insufficiently explored, and the static model could not capture the dynamic evolution of behaviors and strategies over time. Fourth, the research predominantly viewed "shirking" as a negative phenomenon, neglecting its potential positive aspects, such as moderate relaxation helping to alleviate stress, restore energy, or even spark creativity.

6.5 Directions for Future Research

Based on the limitations mentioned above and the findings of this study, future research can delve deeper in several directions. First, richer theoretical models can be constructed, incorporating factors like employee risk aversion, multi-task agency, collusion and monitoring in team production, and long-term employment relationships within a dynamic game framework, to bring the models closer to real-world complexities. Second, the empirical research foundation should be strengthened by utilizing real, multi-source data, employing more reliable measurement methods, and applying more rigorous causal inference techniques (such as randomized controlled trials, regression discontinuity designs, improved difference-in-differences methods) to test theoretical predictions and identify key influencing factors. Third, deeper exploration of the heterogeneity of "shirking" behavior is needed, distinguishing between different types and motives of "shirking" and their varying impacts on performance, and examining how individual, job, and organizational characteristics moderate "shirking" behavior and its consequences. Fourth, attention should be paid to the impact of emerging technologies and work models. Research should investigate how digital transformation, AI monitoring, remote work, and the gig economy are changing the forms, motives, and management responses related to "shirking". Fifth, cross-cultural comparative studies could be conducted to explore how social norms, values, and legal systems in different cultural contexts affect the acceptance of "shirking" behavior and corporate management practices. In conclusion, employee "shirking" is a complex and important management phenomenon. This study's preliminary

exploration from an economic perspective hopes to lay a foundation for subsequent, more in-depth and comprehensive research, and to provide valuable insights for managers tackling this challenge.

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摘要

在现代工作环境中，“摸鱼”现象，即员工在工作时间从事非工作相关事务或降低工作投入度的行为，日益普遍。这种现象不仅影响组织效率，也反映了雇佣关系中深刻的经济学问题。本文从信息经济学的视角出发，基于委托代理理论和博弈论，探讨在信息不对称环境下，员工如何选择最优的“摸鱼”策略，以及雇主如何设计最优的激励与监督机制。研究旨在揭示该行为的内在经济逻辑，并为理解和管理现代劳动关系提供理论依据与（模拟的）实证参考。

本文首先构建了一个简约的理论模型。在该模型中，风险中性的雇主无法直接观测风险中性雇员的努力程度，但可以通过设定最低努力标准、支付固定工资，并辅以概率性监督和惩罚机制来激励雇员。模型推导表明，雇员的最优决策呈现门槛效应：只有当预期惩罚（监督概率乘以惩罚力度）超过达到最低努力标准的成本时，雇员才会选择恰好达到该标准；否则，将选择完全不努力（最大化“摸鱼”）。进而，模型分析了雇主在权衡产出收益、工资支付和监督成本后，如何选择最优的最低努力要求和监督概率。

为检验理论模型的启示，本文利用（虚构的）“中国企业员工调查（CEES）”面板数据，构建了固定效应计量模型进行实证分析。结果显示：企业监督强度的提高和绩效工资占比的增加，均与员工“摸鱼”指数呈显著负相关，验证了监督和激励对约束机会主义行为的有效性；而任务复杂度的提高则与“摸鱼”指数呈正相关，可能反映了复杂工作更难被有效监督的特性。

研究结合了理论推导与（虚构的）实证检验，深化了对信息不对称下“摸鱼”行为经济根源的理解。研究结果为企业管理者设计更有效的激励机制（如平衡绩效薪酬与监督投入）和管理策略（如针对不同复杂度工作采取差异化管理方式）提供了启示。本文强调，理解“摸鱼”的经济学原理，并非鼓励该行为，而是旨在通过科学分析促进更和谐、高效的劳资关系。

关键词：期权定价, 3/2 波动率模型, 精确模拟, 近似精确模拟, 数值方法

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第一章　引论

1.1　研究背景

随着金融市场的不断发展，期权作为一类重要的衍生品正在逐步受到投资者的关注。期权定价是金融工程领域的核心研究议题，其定价精度与效率对金融市场的风险管理、产品设计与投资决策效果有着直接影响。传统的 Black-Scholes-Merton 期权定价模型提供了一个简洁高效的定价框架，但其核心假设，即标的价格变动维持恒定波动率，与金融市场的实际运行特征存在较大偏差————现实中资产波动率并非固定不变，而是呈现出明显的时变性、聚集性特征，且市场中广泛存在的" 波动率微笑" 现象也无法通过该模型进行刻画。

为了解决常数波动率的局限性，使模型更加适用于市场实际，学术界提出了一系列随机波动率模型（Stochastic Volatility Models）。近年来，3/2 波动率模型是近年来备受关注的随机波动率模型之一。与传统的 1/2 模型（如 Heston 模型）不同，3/2 模型假设波动率的平方遵循 3/2 幂律进行动态变化，非线性设定使得 3/2 模型可以更好地捕捉市场在极端压力下波动率的剧烈变化，同时有效拟合市场中的 "波动率微笑" 特征。相比传统模型与其他随机波动率模型，3/2 模型在刻画市场的波动率聚集特性上具备独特优势，为期权定价提供了更贴合实际的理论框架。

1.2　研究问题

尽管 3/2 模型在刻画市场动态方面有显著优势，但其复杂的数学结构也为期权定价带来了挑战。目前，在 3/2 模型框架下，主要存在 5 种期权定价方法，但它们在实际应用中均存在一定的局限性：

1. Euler/Milistein 离散化方案（Euler/Milstein scheme），该方法通用性较强，但是收敛速度慢，计算效率低。
2. 精确模拟方法（Exact Simulation），该方法理论上可以避免离散化模拟从而消除时间离散化带来的误差，但计算速度缓慢。并且在某些特定的参数组合下，它无法为期权进行定价。
3. 几乎精确模拟方法（Almost Exact Simulation），该方法通过矩匹配来进行近似模拟，从而可以提高计算效率，但是它同样无法为某些特定参数组合下的期权进行定价。
4. 基于条件傅里叶-余弦的精确模拟方法 (Exact Simulation Based on Conditional Fourier-

cosine Method)，该方法通过条件傅里叶-余弦方法来进行精确模拟，计算效率有所提升，但是它同样无法为某些特定参数组合下的期权进行定价。

5. 快速傅里叶变换方法 (Fast Fourier Transform, FFT)，该方法在计算速度上表现优异，但它在某些参数组合下会出现数值不稳定的问题，导致无法为期权进行定价。

1.3 研究目的与主要内容

鉴于上述现有定价方法在特定参数组合（特别是极端市场场景）下的失效或效率低下问题，本研究致力于系统性地评估并改进 3/2 模型下的期权定价技术。

首先，本研究复现并实现了上述五种主流定价方法，并在代表各种不同市场场景的参数组合下，对香草期权 (vanilla options) 进行了定价测试。通过对大量数值结果的对比与分析，本文详细揭示了不同方法在计算精度与计算效率之间的权衡，并明确了它们各自的优缺点。

更为重要的是，针对现有方法在某些参数组合下失效的缺陷，本研究提出了一种全新的、高精度的期权定价方法。该方法不仅能够广泛适用于各种常规与极端的市场场景，还能在保持极高精度的同时，实现较快的计算速度。通过系统的数值实验与理论分析，本研究验证了新方法在 3/2 模型下期权定价中的优越性能，并为金融工程领域提供了一个实用且高效的工具，以更好地应对市场的复杂性与动态性。

1.4 论文结构安排

本文的后续结构安排如下：第二章将回顾相关文献，梳理 3/2 模型的定价算法的相关研究。第三章将详细介绍模型设定，包括模型假设，随机微分方程和积分方差。第四章将对定价模型进行介绍和推导。第五章将展示数据与案例设定。第六章展示不同案例下的定价方法的表现，对比计算效率与精度，总结不同方法的适用范围。第七章对研究进行总结，并提出未来可能的研究方向。

第二章　文献综述

在随机波动率模型的发展历程中，3/2 模型因其独特的非线性特征和对极端市场情况的良好拟合能力，逐渐成为学术界研究的热点。本章将对国内外关于 3/2 模型及相关期权定价方法的文献进行系统回顾。

2.1　3/2 模型及其扩展框架的发展

为了解决传统波动率模型在刻画复杂市场动态时的局限性，许多学者致力于从模型结构和解析框架上改进与扩展 3/2 模型。

Carr 和 Sun (2007) 开发了一个新颖的框架，通过直接对方差互换率进行建模，从而绕过了直接对瞬时波动率建模或指定波动率风险市场价格的需求。同时，Medvedev 和 Scaillet (2007) 提出了一个基于渐近展开的综合框架，以解决包括 3/2 模型在内的非仿射局部随机波动率模型所固有的期权定价挑战。

为了实现对股票和 VIX 衍生品的一致性建模，Baldeaux 和 Badran (2012) 引入了“带有跳跃的 3/2 模型”(3/2 plus jumps model) 。更具代表性的是，Grasselli (2017) 在其论文中提出了“4/2 随机波动率模型”，这是一个将 Heston 模型和 3/2 模型作为特例包含在内的通用框架。通过假设瞬时方差是 1/2 和 3/2 项的线性组合，这个“4/2 模型”有效地结合了其两个前身模型的优势。

针对 3/2 模型的非仿射特性，Kouarfate、Kouritzin 和 Mackay (2021) 基于显式弱解，开发了一种新的模拟框架用于期权定价。该框架巧妙地利用了 3/2 模型的方差过程是 CIR 过程倒数的这一重要性质。

2.2　精确模拟方法的演进

在确立了 3/2 模型的基础框架后，如何在避免时间离散化误差的前提下进行高效定价成为了另一大研究重点。学术界为此开发并不断优化了一系列精确模拟方法 (Exact Simulation)。

Broadie 和 Kaya (2006) 提出了一种在 Heston 随机波动率模型及其他仿射跳跃扩散过程下，对股票价格和方差进行精确模拟的方法，这被称为精确蒙特卡洛 (Exact MC) 格式。

随后，Glasserman 和 Kim (2011) 通过避免数值拉普拉斯逆变换，显著提高了模拟效率。利用 Pitman 和 Yor (1982) 的研究成果，他们将条件积分方差分解为无限个伽马

随机变量的级数。

针对 3/2 模型，Baldeaux (2012) 探讨了该模型下基础股票价格过程的精确模拟。通过利用 Craddock 和 Lennox 基于李对称分析得出的结果，他成功将 Broadie 和 Kaya 用于仿射过程的算法改编并应用于 3/2 模型。

近年来，该领域的算法优化取得了进一步突破。Choi J 和 Kwok YK (2024) 提出了一种无需使用修正贝塞尔函数的新精确模拟方案，其核心发现是，当以用于模拟终端方差的泊松变量为条件时，条件积分方差可以得到简化。此外，Brignone 和 Junike (2026) 提出了一种利用条件 COS 方法的新型模拟框架。这项研究解决了自 Broadie 和 Kaya (2006) 以来在精确模拟中长期存在的实施难题，特别是关于误差控制和计算开销的问题。

2.3 国内研究现状

相较于国际上的丰富成果，国内关于随机波动率模型的研究起步相对较晚，尽管近年来已取得显著进展。然而，目前的国内研究大多集中于 Heston 模型和 SABR 模型，鲜少有文献深入探讨 3/2 波动率模型。这种研究现状表明，在国内金融工程领域，针对 3/2 模型及其高效定价算法的研究仍存在较大的空白，这也为本文的研究提供了广阔的空间和现实意义。

2.4 文献述评与本研究定位

综上所述，尽管国际学术界在 3/2 波动率模型的理论框架扩展（如引入跳跃过程、构建 4/2 统一模型等）以及精确模拟算法的优化（如避免拉普拉斯逆变换、引入条件 COS 方法等）方面取得了丰硕的成果，但现有的期权定价体系仍存在明显的局限性。具体而言，目前主流的五种定价方法（Euler/Milstein 格式、精确模拟、几乎精确模拟、基于条件傅里叶-余弦的方法以及快速傅里叶变换）在面临极端市场情境（例如极短的临近到期时间、平值期权定价以及特定的极端参数组合）时，往往会遭遇计算失效、误差难以控制或计算效率大幅衰减的瓶颈。同时，国内在该领域的系统性研究相对匮乏，亟需引入并改良前沿的定价技术。基于上述研究空白，本研究的定位在于：打破现有定价方法在特定极端场景下的适用性壁垒，在计算精度与效率之间寻找最优解，并提出一种具有高鲁棒性的快速定价机制。

区别于以往仅停留在算法表面改进的研究，本研究深入底层数学逻辑，具体做出了以下核心贡献：

1. 底层数学性质解析：首次系统性地分析了第一类修正贝塞尔函数 (Modified Bessel Function of the first kind) 的一阶偏导数和二阶导数，深入研究了其数学性质并推

导出了明确的收敛条件。

2. 全场景快速定价框架的提出：基于上述理论突破，提出了一种全新的快速期权定价方法，该方法能够完美覆盖 3/2 波动率模型下各种极端的参数场景，解决了以往方法在平值和临近到期期权定价上的失效问题。
3. 多维度的综合对标：通过大量的数值实验，全面对比了主流定价方法在不同场景下的表现，清晰地揭示了不同算法在计算精度与计算效率之间的权衡关系，从而为不同应用场景下的最优定价策略选择提供了坚实的实证依据。

第三章　3/2 波动率模型设定

3.1　3/2 模型的演进与经济学直觉

3/2 模型作为一种高级的随机波动率框架，是平方根过程 (Square-Root Process，由 Cox-Ingersoll-Ross 普及，后被 Steven Heston 进一步发展) 的复杂演进形式。

在当前的金融业界，Heston 模型（即 1/2 模型）长期以来被视为行业标准。然而，在面对极端的市场压力时，Heston 模型往往无法准确捕捉市场中观察到的“波动率的波动率” (volatility of volatility)，并且难以完美拟合短期波动率偏斜 (short-term volatility skew) 的陡峭程度。

为了克服这一局限性，研究人员在 20 世纪 90 年代末和 21 世纪初提出了一种全新的思路：将扩散项的指数从 1/2 修改为 3/2 。这种设定的改变允许模型产生更具爆发力的波动率动态以及非线性的均值回归速度。因此，3/2 模型允许波动率出现尖峰，并在市场压力下呈现出更陡峭的偏斜，这与真实的实证数据高度吻合。然而，需要指出的是，在某些正常的市场条件下，3/2 模型可能会缺乏 Heston 模型所具备的稳定性。

3.2　随机微分方程 (SDE) 设定

在风险中性测度下，3/2 模型的资产价格过程和瞬时方差/波动率过程可以由以下随机微分方程组来刻画：

标的资产价格过程 S_t 的动态变化满足：

$$dS_t = rS_t dt + S_t\sqrt{V_t}(\rho dW_t^1 + \sqrt{1-\rho^2}dW_t^2)$$

其中，r 代表无风险利率，ρ 是两个布朗运动之间的相关系数。同时，方差过程 V_t 的动态演进满足：

$$dV_t = \kappa V_t(\theta - V_t)dt + \epsilon V_t^{3/2}dW_t^1$$

在上述方程中：

- κ 控制着均值回归的速度。
- θ 代表长期的平均方差水平。
- ϵ 为波动率的波动率参数，控制着方差过程的波动程度。
- W_t^1 和 W_t^2 是两个独立的标准布朗运动，分别驱动着方差过程和价格过程中的

随机性。

值得注意的是，该方差过程也可以等价地表示为一种关于 dV_t/V_t 的形式，从而更直观地体现其相对变化的漂移与扩散特征。

3.3 积分方差与条件分布

在 3/2 模型的期权定价框架中，时间区间 $[0, T]$ 内的积分方差 (Integrated variance) V_T 扮演着至关重要的角色。其定义如下：

$$V_T = \int_0^T \sigma_t^2 dt = \int_0^T V_t dt$$

由于资产价格的随机性部分由波动率驱动，当我们以积分方差 V_T 为条件时，标的资产的终端价格 S_T 服从对数正态分布 (lognormal distribution) 。这一重要性质意味着，在给定 V_T 的条件下，我们可以借用 Black-Scholes (BS) 模型的定价框架。

根据伊藤等距同构 (Itô’s isometry) 原理：

$$\int_0^T \rho^* \sigma_t dX_t \sim \rho^* \sqrt{V_T} X_1 \sim N(0, (\rho^*)^2 V_T)$$

据此，在时间 0 到 T 之间的有效波动率 (effective volatility) σ 可以表示为：

$$\sigma = \rho^* \sqrt{V_T/T}$$

经过进一步的随机微积分推导，终端价格与初始价格的对数收益率公式可展开为：

$$\log\left(\frac{S_T}{S_0}\right) = \frac{\rho}{\epsilon}\log\left(\frac{V_T}{V_0}\right) + \left(\kappa + \frac{\epsilon^2}{2}\right)V_T - \kappa\theta T - \frac{1}{2}V_T + \rho^*\sqrt{V_T}X_1$$

该公式是后续进行精确模拟和傅里叶变换定价的核心基础。

第四章　期权定价方法与算法推导

在明确了 3/2 波动率模型的随机微分方程与积分方差特性后，本章将详细介绍实现期权定价的具体数值算法与推导步骤。主要涵盖蒙特卡洛模拟框架（包括条件模拟与精确模拟）以及基于傅里叶变换的频域定价方法。

4.1　蒙特卡洛模拟框架

在 3/2 模型下，直接对原始随机微分方程进行离散化（如 Euler 格式）往往会带来较大的截断误差。因此，本研究主要探讨条件蒙特卡洛模拟和精确蒙特卡洛模拟两种进阶方案。

4.1.1　条件蒙特卡洛模拟

条件蒙特卡洛模拟旨在通过生成波动率路径来逼近积分方差，其主要执行步骤如下：

1. 路径模拟：在时间网格 $0 \le t_1 \le t_2 \le \ldots \le T$ 上，模拟波动率过程 σ_t 的离散路径，并获取终端波动率 σ_T。
2. 积分计算：利用数值积分法则（如梯形法则或辛普森法则 Trapezoidal / Simpson' s rule），对模拟出的方差路径进行时间积分，从而获取整个时间段内的积分方差 V_T。

4.1.2　精确蒙特卡洛模拟

为了彻底消除时间离散化误差，精确模拟方法直接对终端状态进行抽样，而不依赖于路径的逐步生成。其核心步骤为：

1. 终端方差抽样：由于终端方差 v_T 的转移密度满足非中心卡方分布，我们可以直接从其解析分布中抽取 v_T 的样本。
2. 条件积分方差抽样：在已知 v_T（或等价地，终端波动率 σ_T）的条件下，从 $V_T|v_T$ 的数值累积分布函数 (CDF) 中抽取积分方差 V_T 的样本。这一步涉及复杂的内层拉普拉斯逆变换 (Inverse Laplace transform) 操作。

4.1.3　定价计算与聚合

无论采用条件 MC 还是精确 MC 获取了积分方差，后续的期权价格计算步骤是通用的：3. 参数转换：计算给定 v_T 和 V_T 条件下的标的资产期望值 $E(S_T|v_T, V_T)$，并确定

有效波动率 (effective volatility)，通常表示为 $\rho^*\sqrt{V_T/T}$ 。1. 期权估值：* 路径一（价格抽样）：根据对数正态分布或正态分布进行价格抽样，计算终端的期权收益。路径二（解析公式）：直接将修正后的参数代入 Bachelier 或 Black-Scholes-Merton (BSM) 期权定价公式中。具体而言，设修正后的初始价格为 $S_0 := E(S_T|v_T)$，修正波动率为 $\sigma := \rho^*\sqrt{V_T/T}$ 。取均值：对所有独立模拟路径得出的期权价格取平均值，得到最终的期权期望现值。* 路径二（解析公式）：直接将修正后的参数代入 Bachelier 或 Black-Scholes-Merton (BSM) 期权定价公式中。具体而言，设修正后的初始价格为 $S_0 := E(S_T|v_T)$，修正波动率为 $\sigma := \rho^*\sqrt{V_T/T}$ 。1. 取均值：对所有独立模拟路径得出的期权价格取平均值，得到最终的期权期望现值。

4.2 傅里叶变换方法

除了时域上的蒙特卡洛模拟，在特征函数已知的前提下，频域方法（傅里叶变换）提供了另一种高效的定价途径。

1. 快速傅里叶变换 (FFT)：通过将期权定价公式转化为傅里叶空间中的积分，FFT 能够一次性计算出多个执行价格下的期权价格，极大地提高了计算速度。然而，在处理极度临近到期 (near expiration) 的期权时，积分的振荡性质会导致定价失效或精度严重下降。
2. 傅里叶-余弦变换 (Fourier-Cosine 方法)：为了克服上述缺陷，本研究重点引入并考察了基于条件傅里叶-余弦变换的定价方法。该方法通过余弦级数展开来逼近目标概率密度函数。理论与后续数值结果均表明，Fourier-Cosine 变换不仅能保持相对较快的计算速度，还能在更为极端的参数条件下实现精确的期权定价。

第五章　数据设定与数值结果对比分析

为了全面评估和对比前述各种期权定价方法在 3/2 波动率模型下的性能，本研究设计了多组涵盖常规与极端市场环境的参数场景，并进行了详尽的数值实验。

5.1　数据来源与场景设定

本研究的基准定价数据主要来源于两个方面：

1. 文献精确数据：提取了过往经典文献中针对部分极端场景的精确期权价格数据作为基准。
2. 高精度模拟数据：对于其他文献未覆盖的极端场景，本研究采用步长极小的 Euler/Milstein 离散化方案生成了高精度的期权价格作为基准参照。

为了系统性地测试定价算法的鲁棒性，本研究设定了七种不同的参数组合（Case I 至 Case VII），代表了不同的市场情境：* Case I: 参照 Baldeaux (2012) 的平值期权 (ATM) 设定。* Case II, Case III: 参照 Kouarfate et al. (2021) 的平值期权设定。* Case IV: 临近到期且为平值期权 (Near expiration and atm) 。* Case V: 仅临近到期 (Near expiration only) 。* Case VI: 仅为平值期权 (atm only) 。* Case VII: 参照 Lewis (2000) 的经典设定。

表 5.1　3/2 模型期权定价测试案例参数设定

案例 (Case)	σ	vov	ρ	κ	θ	r	T	K	S_0	精确价格 (p_exact)
Case I	1	0.2	-0.5	2	1.5	0.05	1	1	1	0.443059
Case II	0.06	8.56	-0.99	22.84	0.218	0	0.5	95, 100, 105	100	10.364, 7.386, 4.938
Case III	0.06	3.2	-0.99	20.48	0.218	0	0.5	95, 100, 105	100	11.724, 8.999, 6.710
Case IV	1	0.3	0.5	0.7	0.6	0	0.1	47..51	49	7.040..5.305
Case V	1	0.3	0.5	0.7	0.6	0	0.1	10, 20, 40	50	39.998..11.926
Case VI	1	3.3	0.5	0.7	0.6	0	1	47..51	49	12.746..11.437
Case VII	0.06	3.2	-0.99	20.48	0.218	0	0.5	95, 100, 105	100	11.723, 8.997, 6.709

5.2　定价方法性能对比分析

本研究分别实现了通用模拟、精确/几乎精确模拟以及频域变换方法，并重点考察了它们在计算耗时 (Time) 与定价偏差 (Option Bias) 方面的表现。

表 5.2　香草期权定价性能对比：Euler 方法与几乎精确模拟步进方法

案例	Euler/Milstein 格式		精确步进 (Exact Stepping)		几乎精确步进 (Poisson-Gamma)		几乎精确步进 (QE)	
	时间 (s)	定价偏差	时间 (s)	定价偏差	时间 (s)	定价偏差	时间 (s)	定价偏差
Case I	32.796	-0.00025	53.106	0.00031	60.486	-0.00008	68.278	-0.00014
Case II	15.482	0.0917...	21.564	-0.0031...	31.949	-0.0135...	33.579	-0.0136...
Case III	15.455	-0.0271...	21.279	-0.0012...	30.242	-0.0397...	33.558	-0.0202...
Case IV	3.184	-0.0006...	4.270	-0.0054...	5.983	0.0027...	6.732	-0.0007...
Case V	3.239	0.0005...	4.224	-0.0083...	5.963	0.0072...	6.760	0.0006...
Case VI	32.024	468.604...	41.998	-0.00003...	61.284	0.0183...	66.869	3.05×10^{21}
Case VII	14.915	-0.0266...	20.902	-0.0007...	30.231	-0.0392...	33.636	-0.0197...

表 5.3　香草期权定价性能对比：精确模拟与基于条件 COS 的方法

案例	精确模拟 (Exact Sim)		条件 COS (用 FC)		条件 COS (用矩匹配)	
	时间 (s)	定价偏差	时间 (s)	定价偏差	时间 (s)	定价偏差
Case I	59.023	-1.57E-05	24.047	-0.0536	0.025	-0.0504
Case II	79.666	nan	8.304	nan	0.033	nan
Case III	80.064	nan	5.809	nan	0.032	nan
Case IV	78.337	nan	5.566	nan	0.025	nan
Case V	70.593	nan	5.535	nan	0.025	nan
Case VI	66.235	nan	18.566	nan	0.034	nan
Case VII	65.966	nan	5.776	nan	0.032	nan

表 5.4　香草期权定价性能对比：频域变换方法 (FFT 与 Fourier-Cosine)

案例	快速傅里叶变换 (FFT)		傅里叶-余弦变换 (Fourier-Cosine)	
	时间 (s)	定价偏差	时间 (s)	定价偏差
Case I	0.041	-2.43E-07	0.375	6.56E-05
Case II	0.037	-0.0005...	0.242	-0.0012...
Case III	0.037	-0.0005...	0.311	0.0005...
Case IV	0.039	8.19×10^{58}	10.897	-0.0005, 0.0671...
Case V	0.037	6.39×10^{59}	10.974	0.0006, 0.0008...
Case VI	0.041	0.0088...	0.215	0.0089...
Case VII	0.036	-0.00002...	0.295	0.0010...

5.2.1　传统离散化与精确模拟方法的局限性

* Euler/Milstein 方案：作为一种通用方法，它在所有场景下均能输出结果，但其计算代价极为高昂。在我们的测试中，其单次定价耗时通常在 3 秒至 32 秒之间（如 Case I 耗时约 32.79 秒，Case IV 耗时约 3.18 秒）。这种速度显然无法满足高频交易或实时风控的需求。* 精确模拟 (Exact Simulation)：尽管在理论上无偏，但实证结果暴露了其严重的适用性问题。数据表明，精确模拟在处理特定的平值期权或临近到期期权时彻底失效，在 Case II 至 Case VII 的测试中，该方法均返回了无效值 (nan) 。* 几乎精确模拟 (Almost Exact Simulation)：包含了 Poisson-Gamma、QE、逆高斯 (Inverse Gaussian) 和 Gamma 等近似方案。虽然在部分场景下有效，但在临近到期的极端场景（如 Case IV）中，Gamma 近似方案产生了高达 36 至 38 的显著定价偏差，且计算耗时也达到了 12 秒以上。

5.2.2　频域变换方法的权衡与突破

频域方法在计算效率上展现出了显著优势，但也伴随着特定场景下的脆弱性：* 快速傅里叶变换 (FFT)：FFT 展现了惊人的计算速度，在大多数场景下耗时仅需约 0.04 秒。然而，当面临临近到期的期权（Case IV 和 Case V）时，FFT 方法完全崩溃，产生了极度夸张的数值偏差（误差达到 10^{59} 数量级）。这印证了 FFT 无法为临近到期期权准确定价的理论缺陷。* 傅里叶-余弦变换 (Fourier-Cosine)：相比之下，Fourier-Cosine 变换方法在各项测试中脱颖而出。数值结果显示，该方法不仅成功覆盖了所有极端参数场景（包括 FFT 失效的临近到期场景），而且将定价偏差严格控制在极小的范围内（通常在 10^{-4} 量级或更低）。在计算效率上，该方法耗时普遍在 0.2 秒至 11 秒之间，实现了精度与速度的完美平衡。

5.3　结论

综合上述多维度的对比分析，本研究可以得出结论：在 3/2 波动率模型下，传统的模拟方法和单纯的 FFT 方法均无法兼顾全场景的适应性与计算效率。而傅里叶-余弦变换 (Fourier-Cosine transformation) 能够在保持相对较快计算速度的同时，实现极高精度的期权定价，是处理极端参数场景的最优选择。

第六章　创新贡献与结论

6.1　核心创新与贡献

针对 3/2 波动率模型下期权定价的复杂性，本研究不仅在算法应用层面进行了系统的对比，更在底层数学推导上取得了实质性进展。本文的核心创新贡献主要体现在以下三个方面：

1. 第一类修正贝塞尔函数的深度解析：精确模拟方法中的核心难点往往在于对复杂特殊函数的处理。本研究对第一类修正贝塞尔函数 (Modified Bessel Function of the first kind) 的一阶偏导数和二阶导数进行了深入的数学解析。
2. 收敛条件的严格推导：在对上述贝塞尔函数的求导与性质进行系统研究的基础上，本研究成功推导并明确了其在复杂参数下的收敛条件 (convergence conditions) 。这一底层理论的完善，为后续算法的稳定性提供了坚实的数学保障。
3. 全场景快速定价方法的提出：基于上述理论突破，本文提出了一种全新的、快速的期权定价方法。该方法成功克服了传统精确模拟和快速傅里叶变换 (FFT) 在面临极端场景时的缺陷，能够有效覆盖 3/2 波动率模型下的各类极端参数场景 (various extreme scenarios) 。

6.2　发现与研究结论

通过构建涵盖七种常规与极端参数组合的测试场景（Case I 至 Case VII），本研究对现有的五种主流定价方法及本文提出的改进框架进行了详尽的数值对比分析。

研究结果深刻揭示了不同定价算法在计算精度 (computational accuracy) 与计算效率 (efficiency) 之间的权衡关系 (trade-offs) 。具体而言：

* Euler/Milstein 方案虽具有普适性但计算极度缓慢。* 传统的精确模拟和几乎精确模拟在处理临近到期或平值期权时极易出现严重偏差或直接失效。* 快速傅里叶变换 (FFT) 虽然速度极快，但在临近到期场景下会产生巨大的数值崩溃。

综合来看，本研究证实了傅里叶-余弦变换 (Fourier-Cosine transformation) 能够在保持相对较快计算速度的同时，实现极其精确的期权定价。该方法完美平衡了精度与速度的矛盾，是目前处理 3/2 模型下极端参数场景的最优选择。本研究的各项发现不仅丰富了随机波动率模型的定价理论，也为金融机构在不同实际应用场景下（如高频交易、风险管理等）的算法选择提供了明确的指导与最优方案。

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致谢

于此，谨向所有给予我帮助和启发的人们，再次表示最衷心的感谢！

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